AI-POWERED HEALTHCARE AGE PREDICTION

Using Salesforce + ML

TEAM MEMBERS :-

T. ARYAMAN RAJ - 22911A3503

K. CHANDRA SHEKAR - 22911A3524

Y. ABHILASH REDDY - 22911A3563

BOLLAJU KEERTHI - 22911A3572

DASARI PRANAVA - 22911A3575

KAVALI SNEHITHA - 22911A3590

USE CASE

Accurately predicting an individual's age based on various data inputs is a challenging task due to the inherent variability and complexity of the data.

Traditional methods of age verification can be intrusive, time-consuming, and prone to errors. Inadequate age prediction can lead to non-compliance with regulations, suboptimal personalization, and security vulnerabilities.

Age prediction involves estimating an individual's age using various data inputs, such as images, text, voice, or other biometric information.

INTRODUCTION

Artificial Intelligence (AI) is transforming healthcare by enabling smarter, faster, and more personalized care. One emerging application is age prediction, where AI models analyze medical data—such as symptoms, lifestyle habits, and biometric indicators—to estimate a patient's biological age. This helps doctors detect age-related health risks early, tailor treatments more precisely, and improve overall patient outcomes. By combining machine learning with real-world health data, AI-powered age prediction offers a powerful tool for preventive medicine and personalized care.

OBJECTIVE

Build an AI-powered system in Salesforce to predictpatient age quickly, securely, and accurately.

KEY GOALS:
Automate age prediction from photo & details, integratewith Salesforce (LWC + custom object), deploy MLmodel as API (Apex), and provide doctors with secureaccess.
Display results on Lightning Record Page

DIFFERENT METHODS OF APPROCHES

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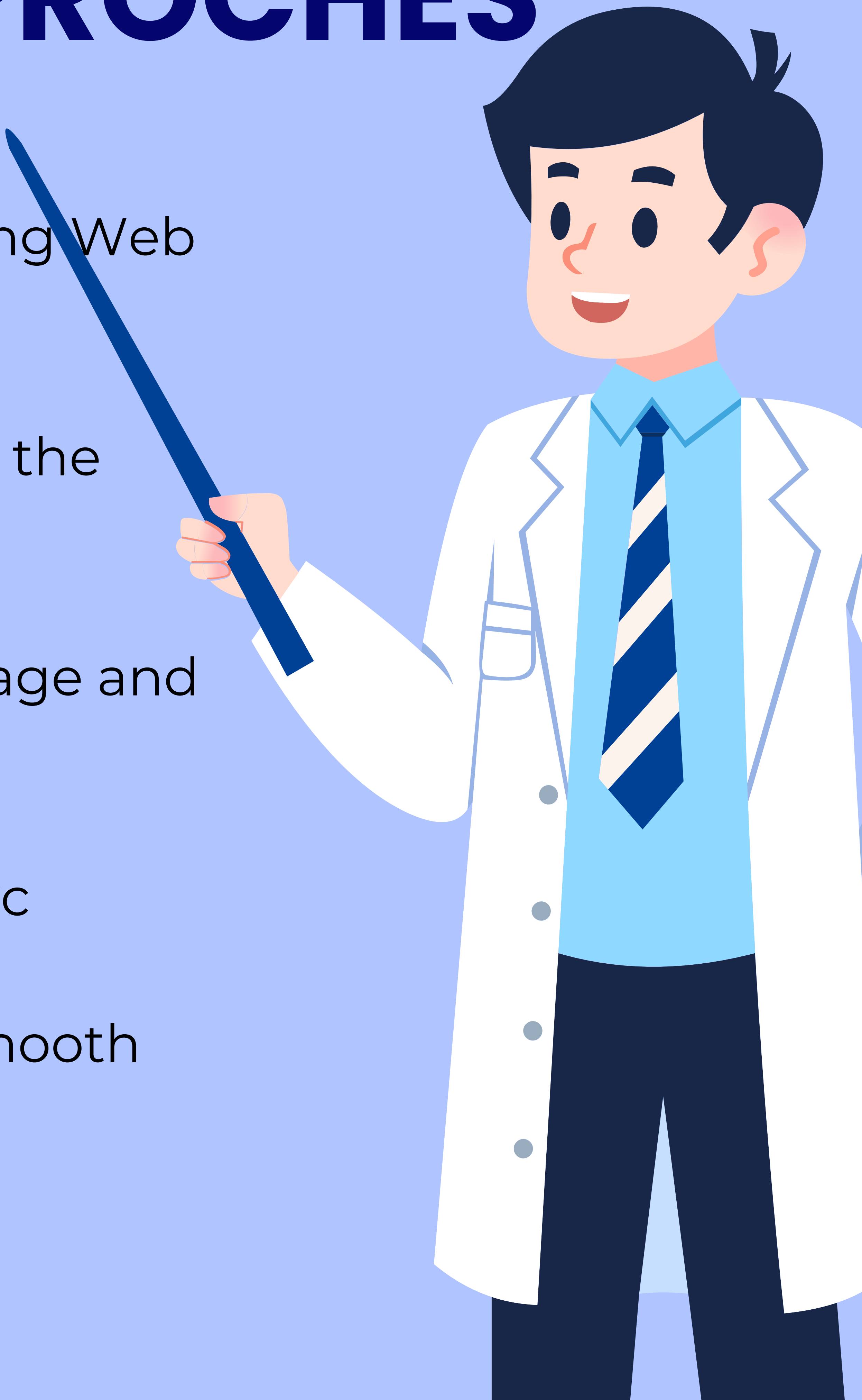
STEP 1: User uploads an image through a Lightning Web Component (LWC) with basic validations.

STEP 2: Apex trigger captures the upload → sends the image (base64) to the Flask API.

STEP 3: The VGG16 CNN model processes the image and predicts age & gender (~81% accuracy).

STEP 4: Predicted Output will display through lwc

STEP 5: Verified with sample images to ensure smooth end-to-end integration.



02

STEP 1: Train Model – Use Kaggle dataset → Python ML libraries → export trained model.

STEP 2: Deploy Model – Host on cloud (AWS, GCP, Heroku) → expose REST API.

STEP 3: Register in Salesforce – Use Einstein Studio (BYOM) → connect API → set auth + input mapping.

STEP 4: Create Custom Object – "Age Prediction Record" with fields (input, predicted age, confidence, timestamp).

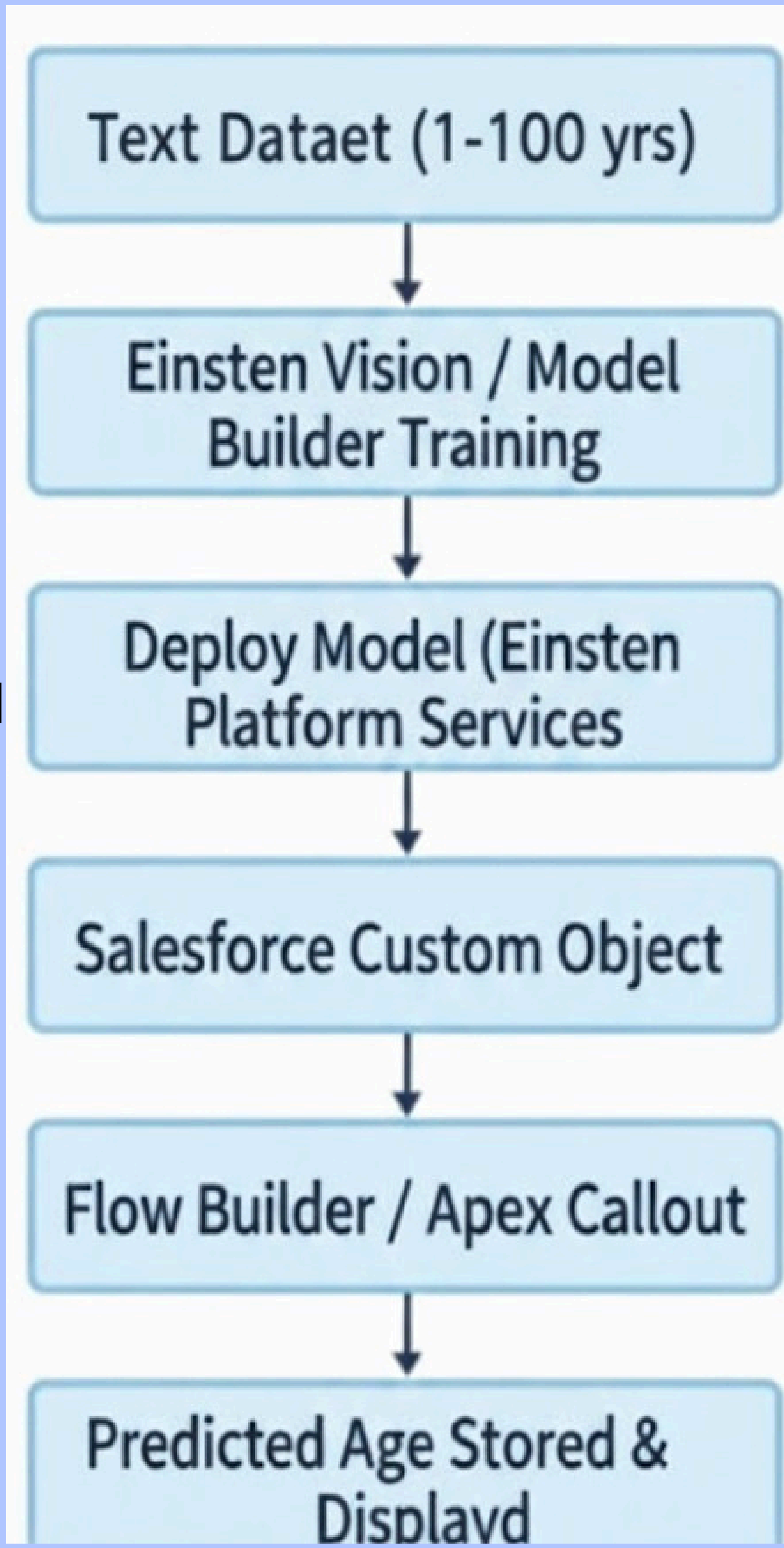
Step 5: Automate with Flow – Trigger on record → call external API → update fields with prediction.

STEP 6: Build LWC UI – Upload input (image/text) → create record → display prediction.

STEP 7: Add to Page – Place LWC in Lightning App Builder for easy access.

STEP 8: Test End-to-End – Upload test data → verify predictions populate correctly.

STEP 9: Monitor & Maintain – Track API usage, update model, show analytics in dashboards.



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STEP 1: Collect Image Dataset (Image Dataset)

Gather labeled images of individuals aged 1–100 years.

Input Format: JPG or PNG with correct age labels.

STEP 2 : Upload Dataset (Einstein Model Builder)

Upload images to Einstein Vision / Model Builder for training.

STEP 3: Configure Model (Einstein Model Builder)

Set parameters: image size, augmentation, number of epochs, etc.

STEP 4: Train Model (Einstein Model Builder)

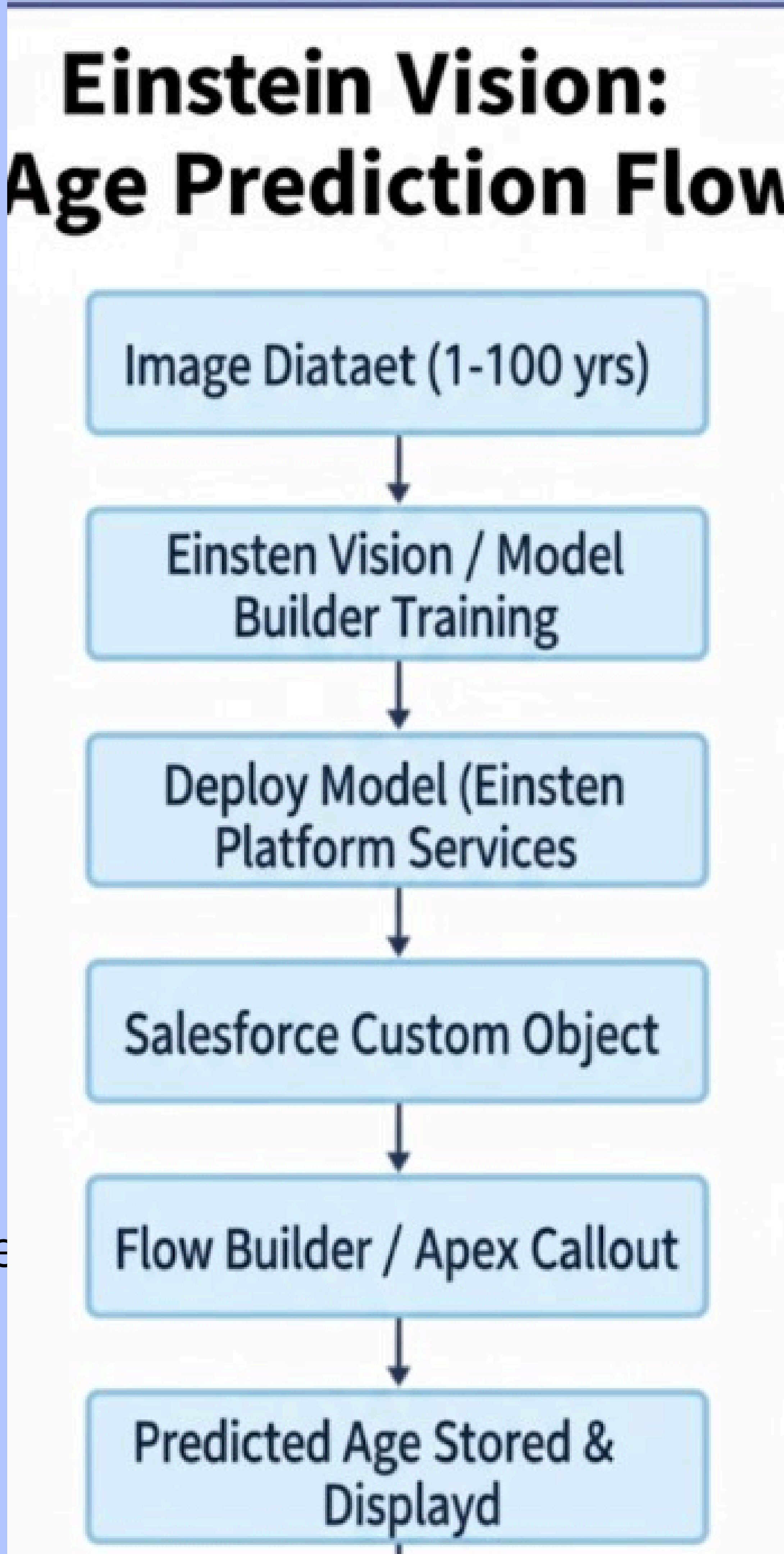
Model learns to predict age from images.

STEP 5: Deploy Model (Einstein Platform Services)

Deploy the trained model as a REST API endpoint.

STEP 6: Upload Input (LWC / Record Page)

Users upload patient images via Lightning Web Compone (LWC) or Salesforce record page.Ensure images are in supported formats (JPG/PNG).



STEP 7: Integration (Flow Builder / Apex Callout)

Flow or Apex sends uploaded images to the deployed Einstein Vision API.Receives predicted age and confidence in JSON.

Step 8: Store Prediction (Salesforce Custom Object)

Save predicted age, confidence, and timestamp in a Custom Object for tracking.

STEP 9: Display & Insights (Reports / Dashboards / Automation) Show predictions in Salesforce record pages, reports, dashboards, and automate workflows.

PROPOSED SOLUTION

STEP 1: Image Upload via LWC

Doctors upload images through a custom LWC.

The LWC provides instant feedback and shows a placeholder while the image is being processed.

Handles client-side validation .

Sends the image to Salesforce backend (Apex) for API callout.

STEP 2: SETUP NAMED CREDENTIALS FOR API INTEGRATION

In Salesforce, configure a Named Credentials that store API URL and authentication securly. Avoids hardcoding endpoints in Apex

STEP 3: Apex Trigger & API Callout

Apex Trigger captures image upload events on the object.

Apex class sends the image to the deployed Flask API (https://salesforce-age-gender-api.onrender.com/predict) using HTTP callout.

Image is base64-encoded and sent as JSON payload.

Parses API response (age and gender predictions).

STEP 4: Salesforce Record Update

Apex updates record fields: Predicted_Age__c Predicted_Gender__c

Enables workflows, reports, dashboards to use predicted values automatically.

STEP 5: Model & API (Supportive Backend)

Backend uses VGG16 CNN model for age/gender prediction.

Hosted as a Flask API on Render (requires paid instance for heavy model)

Preprocessing: face detection, resizing, normalization ensures accurate predictions (~81% accuracy).

STEP 6 : End-to-End Integration Workflow:

User uploads image via LWC → Apex trigger fires → API callout → Model predicts age/gender → Salesforce record updated automatically

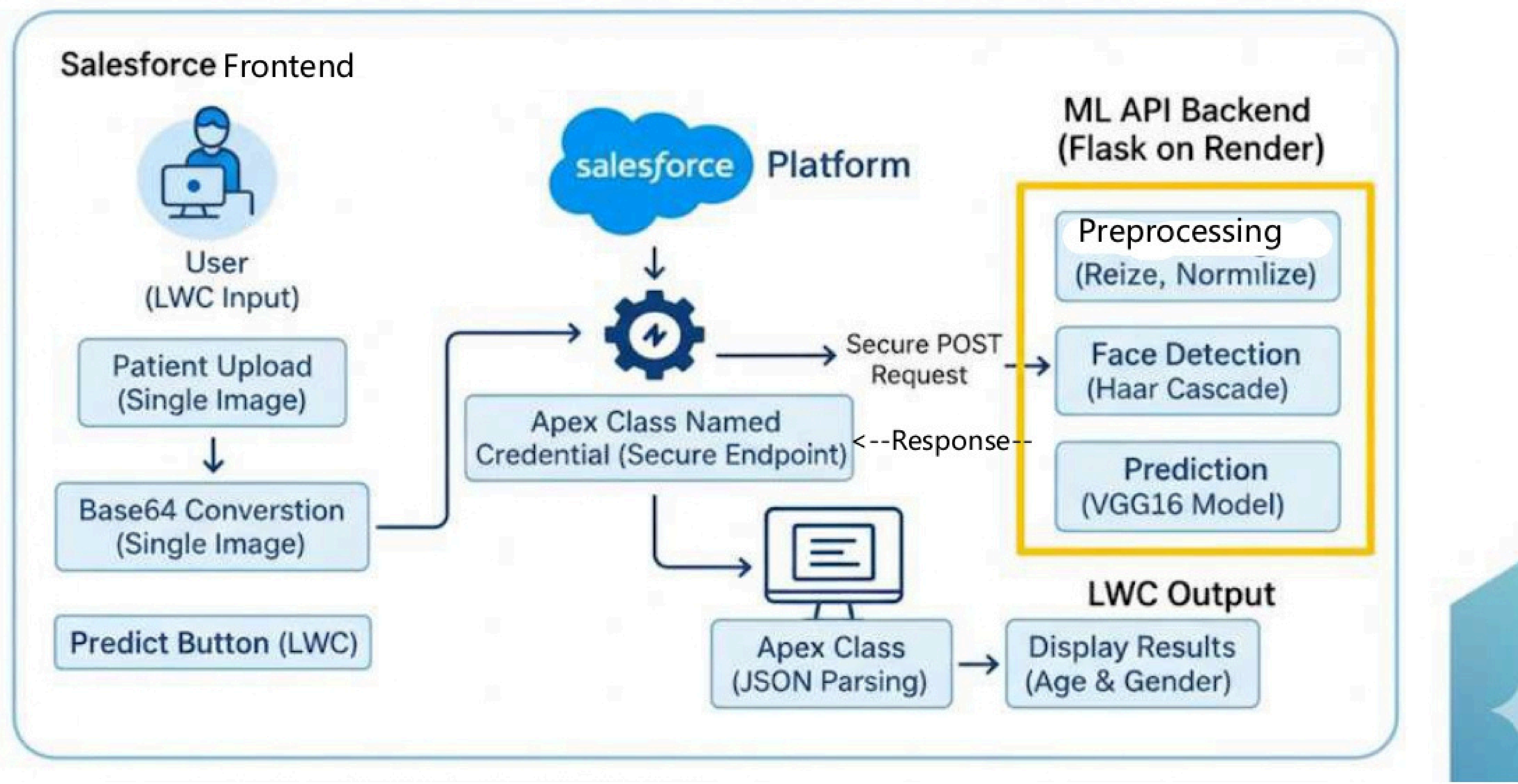
STEP 7 : Testing & Validation

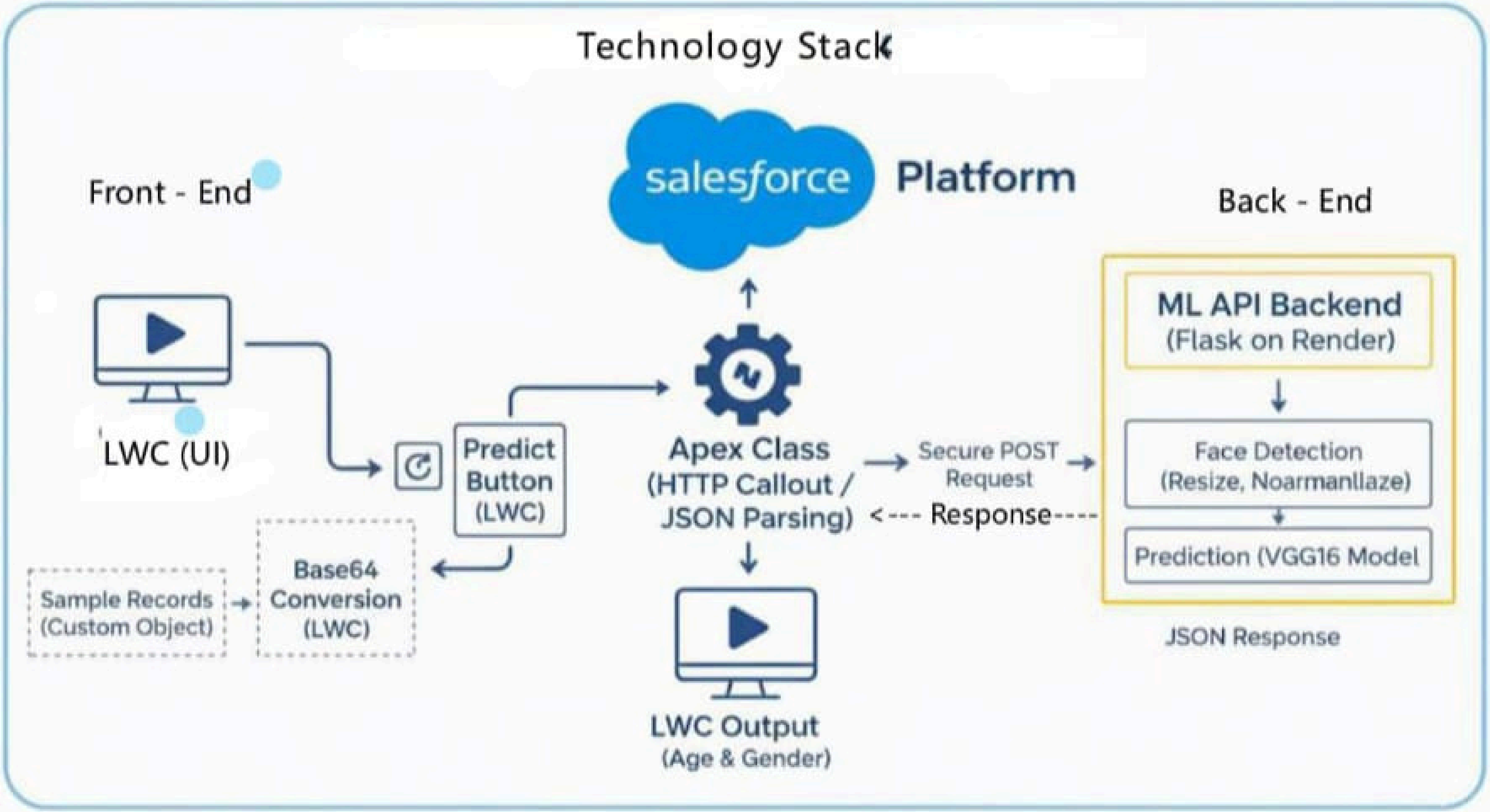
Verified API callouts via Apex and sample images.

Ensured response maps correctly to Salesforce fields.

Showcased end-to-end workflow conceptually; full-scale execution requires paid Render instance.

Architecture Diagram




Technology Stack
salesforce
Platform
Front - End
Back - End
LWC (UI)
Predict Button (LWC)
Apex Class (HTTP Callout / JSON Parsing)
Secure POST Request
<--- Response----
ML API Backend (Flask on Render)
Face Detection (Resize, Noarmanllaze)
Prediction (VGG16 Model
JSON Response
Base64 Conversion (LWC)
Sample Records (Custom Object)
LWC Output (Age & Gender)

FRONTEND

In the frontend, we first created a custom Salesforce app to serve as the interface for the age prediction feature. Within this app, we built Lightning Web Component templates to display the predicted age of the person clearly and intuitively. The predicted age is then displayed dynamically on the templates, ensuring a seamless and user-friendly experience. This setup provides an interactive and visually organized interface for viewing predictions and managing patient records.

BACKEND

On the backend, The Flask/FastAPI API receives the uploaded image from Salesforce in Base64 or multipart format. The image is then preprocessed—converted into an array, resized to the model's input dimensions, and normalized. The CNN (VGG16) model takes this preprocessed image and predicts the age, which is then formatted as a JSON response (e.g., { "predicted_age": 25 }, {"predicted Gender": Male}) and sent back to Salesforce. This ensures the frontend receives a ready-to-use prediction seamlessly.

FRONTEND WORKFLOW

1. APP CREATION

We first created a custom Salesforce app to serve as the user interface for the age prediction feature. This app acts as the central workspace where users can access all related functionalities.

2. BUILDING TEMPLATES

Within the app, we built Lightning Web Component (LWC) templates to display the predicted age in a clear and interactive manner. These templates ensure a user-friendly layout for input and output data.

3. DOCTOR INPUT AND INTERACTION

Doctor can enter patient details or upload images directly through the app interface. The frontend captures these inputs and triggers the backend process to call the external API for prediction.

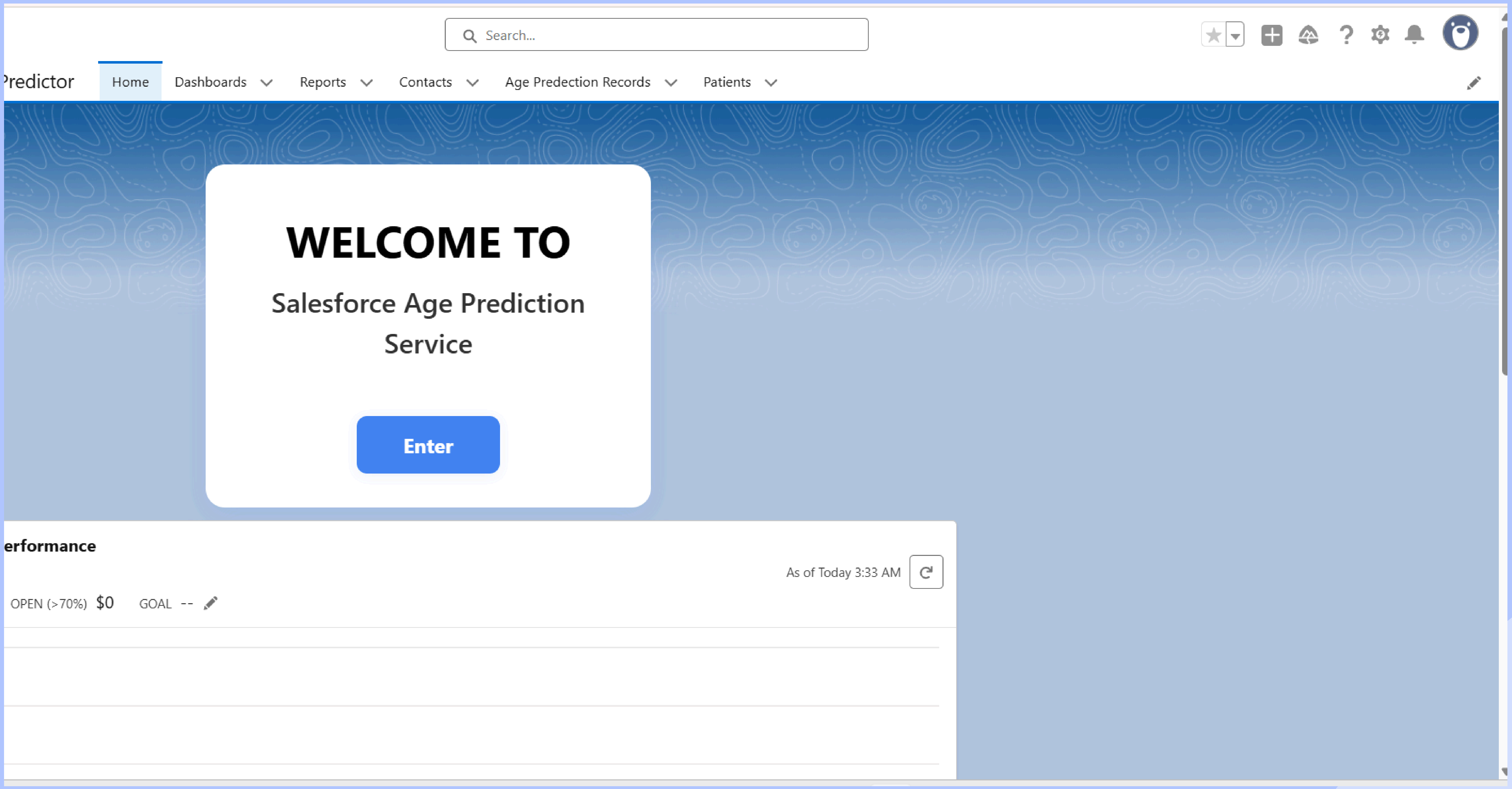
4. Displaying Results

Once the backend returns the predicted age, the LWC templates dynamically display the result on the record page.Additional information such as confidence scores or timestamps can also be shown for better insights.

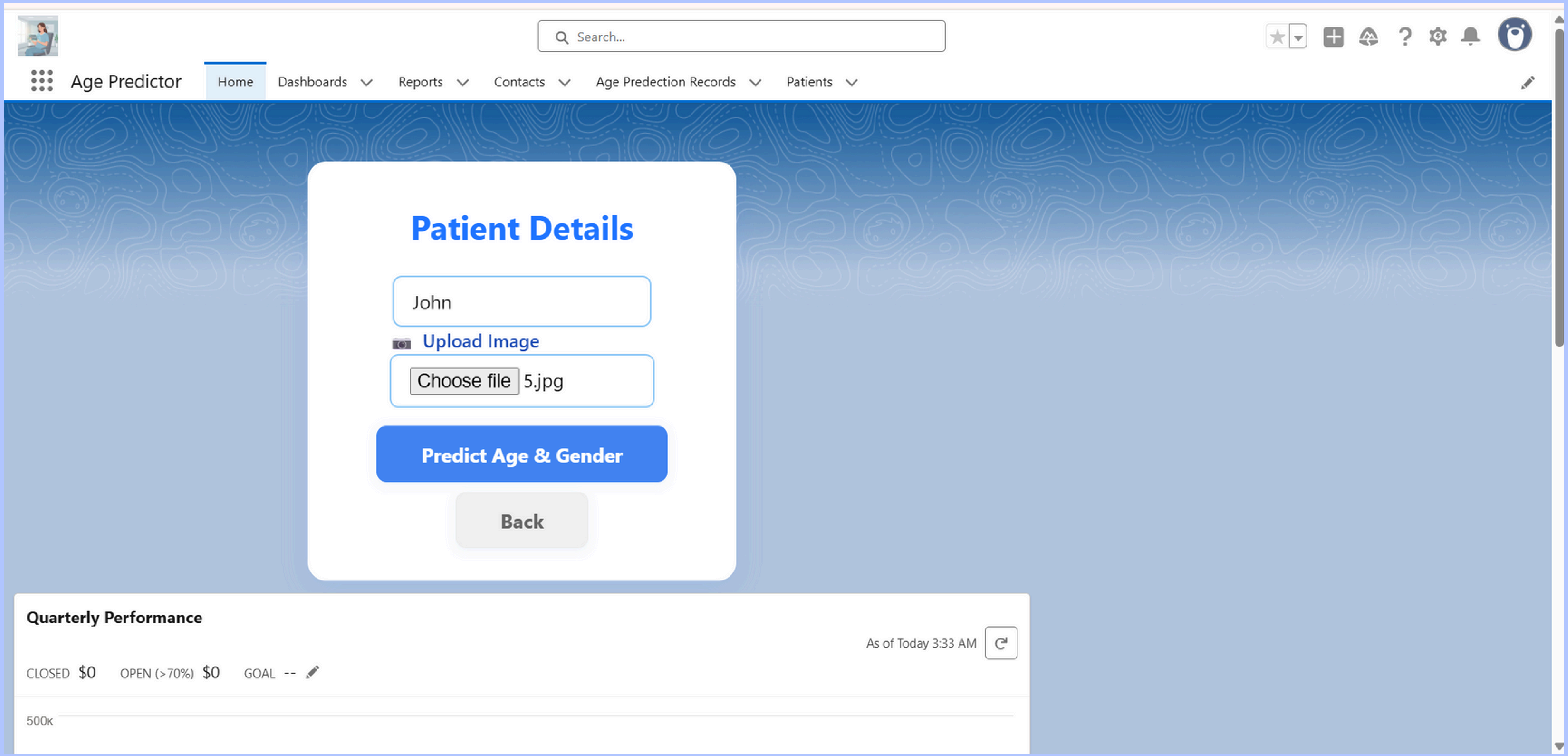
5. SEAMLESS USER EXPERIENCE

The frontend ensures smooth interaction between the user and the system. By integrating forms, templates, and dynamic display components, the solution provides a visually organized and efficient workflow for viewing age predictions and managing patient records.

OUTPUT OF FORNTEND WORK



TEMPLATE 1


Search...
Age Predictor
Home
Dashboards
Reports
Contacts
Age Predection Records
Patients
Patient Details
John
Upload Image
Choose file 5.jpg
Predict Age & Gender
Back
Quarterly Performance
As of Today 3:33 AM
CLOSED $0
OPEN (>70%) $0
GOAL --
500k

TEMPLATE 2

BACKEND WORKFLOW

1. DATA COLLECTION & PREPROCESSING

Collect healthcare or image datasets containing age and gender information.
Clean and preprocess the data to ensure quality for training.
(UTK - Face Dataset (20,000 images, 0 - 116 ages)

2. MODEL TRAINING

Train the machine learning model using Python ML libraries in Google Colab or VS code. The model learns to predict age and gender from images or other data inputs VGG16 (Transfer Learning) Image Preprocessing is done through CV.

3. DEPLOY MODEL VIA RENDER

Upload the trained model to Render. Deploy it as a REST API endpoint with /predict that returns predictions in JSON format, e.g., { "age": 37, "gender": "Male" }.

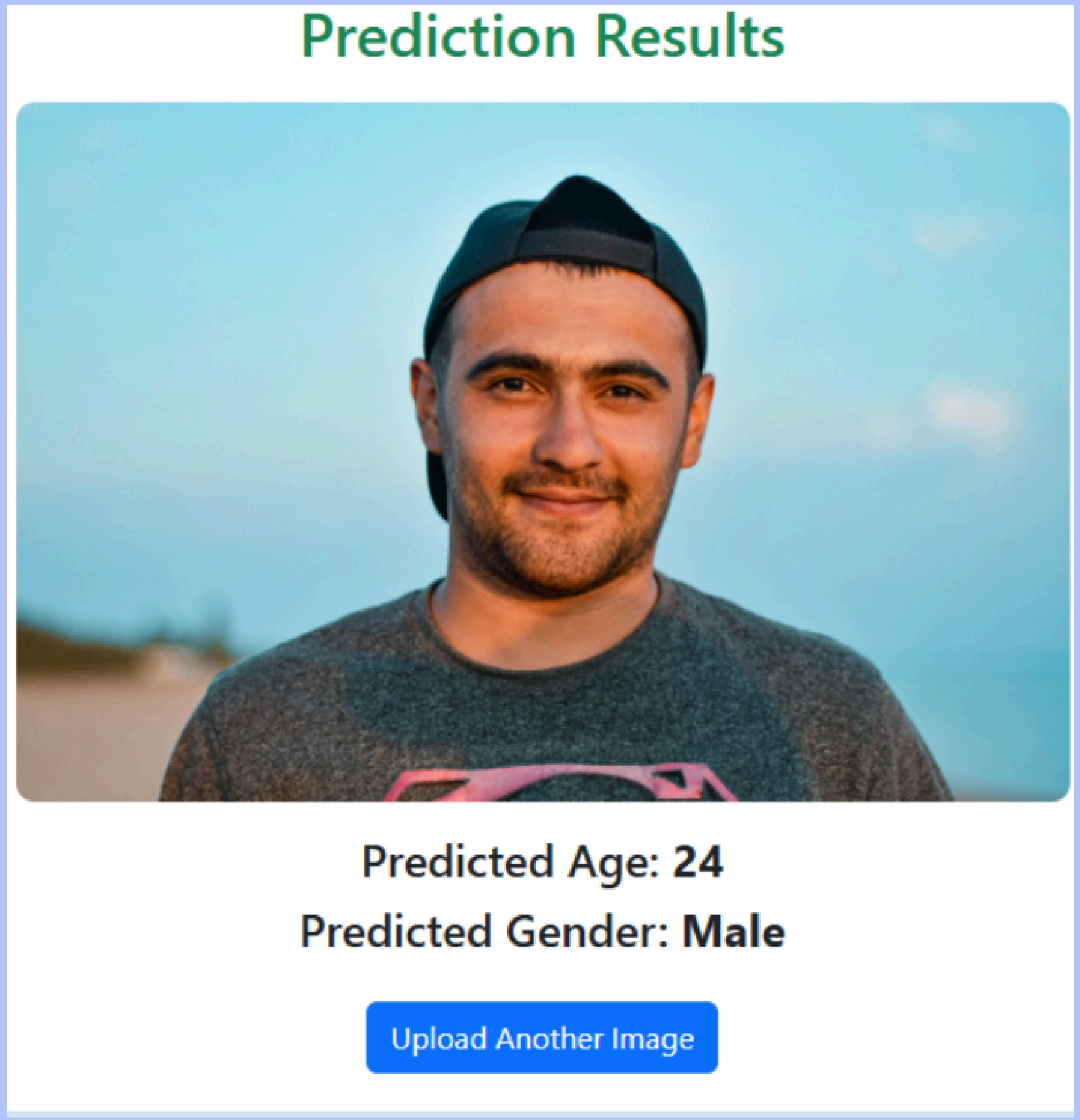
4. SALESFORCE INTEGRATION

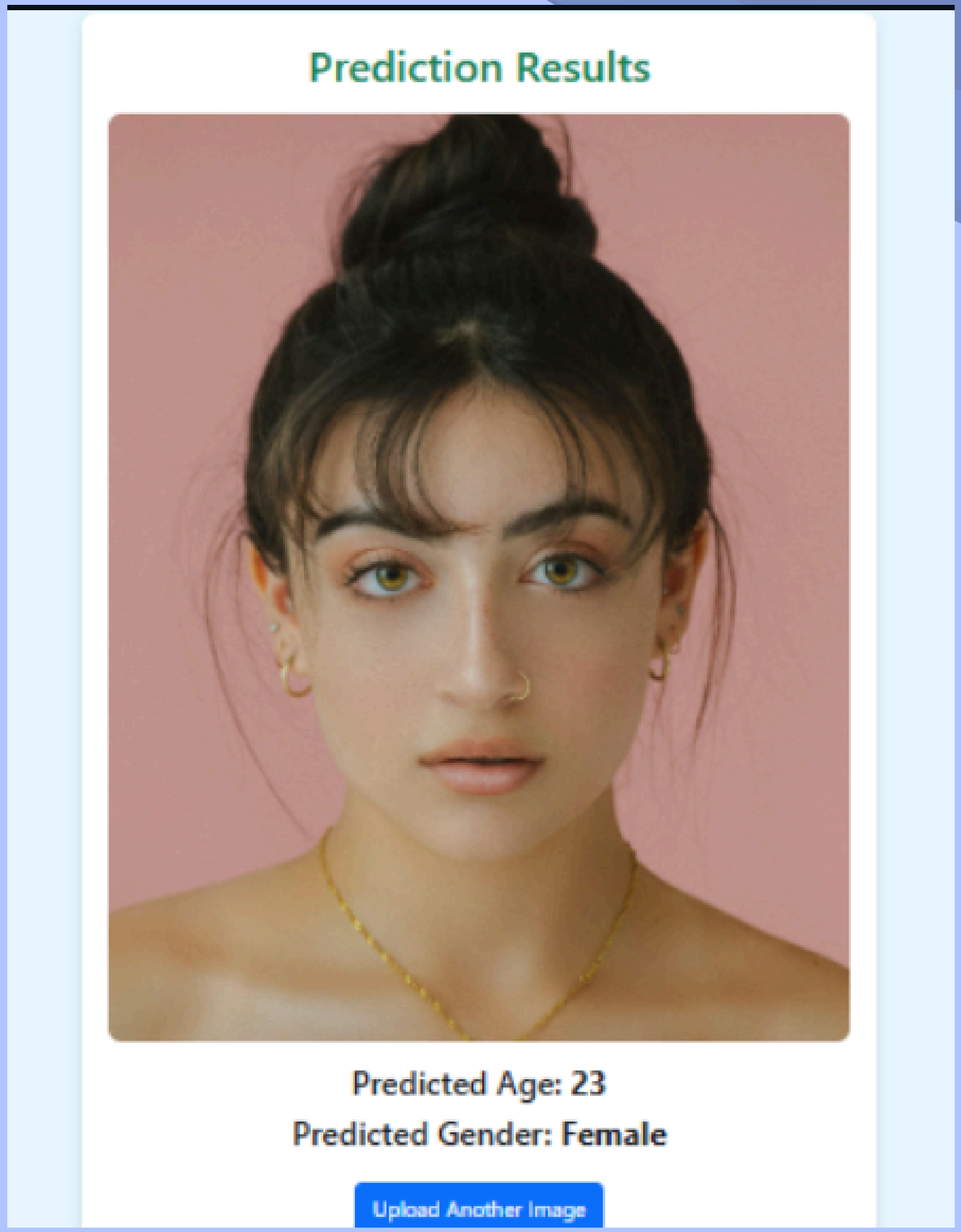
Configure Named Credentials in Salesforce for API authentication. Register the Render API as an External Service. Build a Flow lwc that sends patient data to the API and retrieves the predicted age and gender through Apex Callouts .

5. STORING & DISPLAYING PREDICTIONS

Store the returned predictions in a custom object, e.g., Age Prediction Request,including age, gender, and timestamp. These predictions are then displayed dynamically in the frontend Lightning Web Componentsaragraph text

OUTPUT OF BACKEND WORK





STATISTICS

81%

ACCURACCY

85%

PRECISION

INTEGRATION

STEP 1: Backend API

Model hosted on Flask/FastAPI. Exposes /predict endpoint returning JSON with predicted age.

STEP 2:

Salesforce – Named Credential Stores API URL and authentication securely. Avoids hardcoding endpoints in Apex

Salesforce – Apex Callout , Sends Base64 image from LWC to API.

Parses JSON response and extracts predicted age. Acts as middleware between frontend and backend.

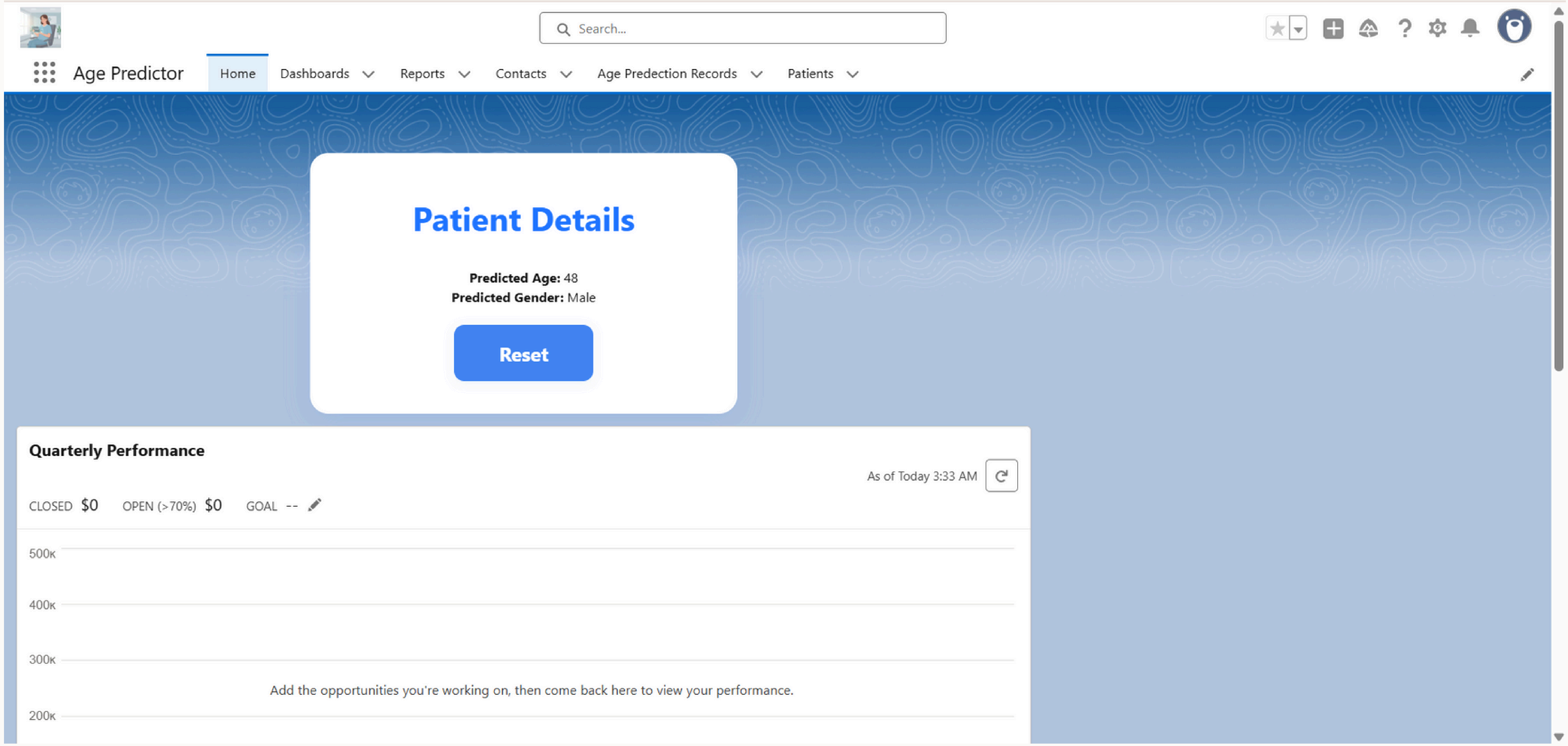
STEP 3: LWC (Frontend)

User uploads image in LWC.LWC → calls Apex method.

Apex → API (Render) → gets prediction → returns to LWC.

LWC displays Predicted Age to user.

OUTPUT AFTER INTEGRATION



LIMITATIONS

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1)Latency Issues – Image conversion to base64 and API callouts may introduce delays, especially with large images or poor network conditions.

2)Scalability Challenges – Current setup may struggle with high volumes of equests in production without a stronger cloud deployment (AWS SageMaker, GCP, etc.).

02

1)Einstein Studio Constraints – Limited flexibility in customization and performance tuning once API is registered in Salesforce.

2)Accuracy Challenges – Predictions may not always be reliable for critical use cases without continuous retraining and monitoring.

3)Maintenance Overhead – Continuous monitoring, retraining, and API updates are required to maintain performance.

03

1)**Accuracy Concerns** – A single CNN model may not provide consistent accuracy across diverse demographics or image qualities.

2)**Security & Compliance** – Transferring patient images outside Salesforce raises data privacy and HIPAA/GDPR compliance risks.

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FUTURE ENHANCEMENTS

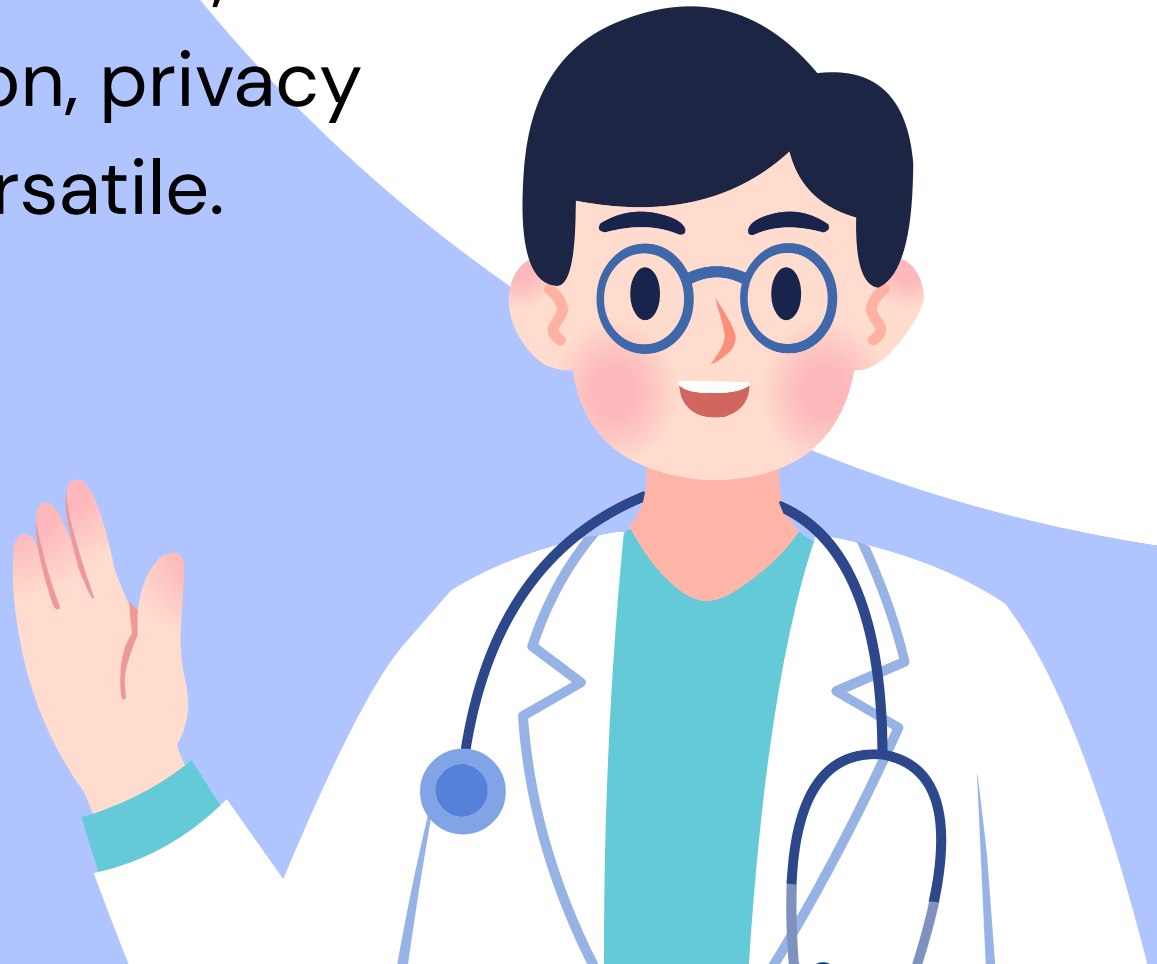
1. **Multi-Modal Inputs** – Extend beyond images to include text, voice, and biometric data for moreaccurate age estimation.
2. **Improved Accuracy with Advanced Models** – Leverage transformers and deep learning architectures to reduce error rates.
3. **Real-Time Predictions** – Enable streaming predictions for live applications such as telemedicine and online onboarding.
4. **Enhanced Security & Complianc**e – Add stronger encryption, consent management, and ensure GDPR/HIPAA compliance.
5. **Scalable Cloud Deployment** – Deploy models on AWS SageMaker, GCP Vertex AI, or Azure ML for large-scale use.
6. **Personalization & Recommendations** – Use predicted age to provide personalized services, product recommendations, or targeted healthcare plans.
7. **Analytics & Dashboards** – Build real-time dashboards in Salesforce to monitor predictions, accuracy trends, and usage metrics.

APPLICATIONS

1.Healthcare & Wellness
2. Security & Access Control
3. Retail & Marketing
4. Forensics & Law Enforcement
5. Social Media & Apps
6. Human-Computer Interaction (HCI)
7. Automotive & Safety Systems
8. Fitness & Lifestyle

CONCLUSION

The AI-powered age detection project successfully demonstrates how Meachine learning and computer vision can estimate human age from facial images with accuracy and efficiency. By integrating the model with cloud APIs and Salesforce, the system showcases practical real-world applications in healthcare, retail, and security. The project highlights the importance of diverse datasets, model optimization, and secure data handling, while future enhancements like real-time detection, privacy protection, and explainable AI can make it even more robust and versatile.



THANK YOU

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